

Secure File Storage System Using Encryption and Access Control

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Abstract

Abstract — In the modern digital world, protecting sensitive data from unauthorized access has become one of the most critical challenges in information technology. Traditional file storage systems often store data in plain text, making them vulnerable to data breaches and unauthorized access. This paper presents a Secure File Storage System that combines Advanced Encryption Standard (AES) encryption with robust user authentication and access control mechanisms. The proposed system ensures that all files uploaded by users are encrypted before being stored on the server, and only authenticated users can retrieve and decrypt their own files. The system is implemented using the Python programming language and the Django web framework. Testing demonstrates that the system effectively protects sensitive data while remaining practical and usable for real-world deployment. This project also contributes to the practical understanding of cryptographic concepts including encryption, authentication, and secure data storage.

Keywords — AES Encryption, Secure File Storage, Access Control, User Authentication, Django, Python, Data Security, Cryptography.

I. Introduction

In the modern digital world, individuals, organizations, and governments generate and store enormous amounts of digital data every day. This data may include personal documents, financial information, research materials, confidential business files, and other sensitive records. With the rapid growth of the internet and cloud technologies, storing and accessing data digitally has become extremely convenient. However, this convenience has also introduced serious security challenges. Cyber threats such as hacking, unauthorized access, data breaches, and identity theft have increased significantly in recent years. As a result, protecting sensitive data has become a major concern in the field of information technology.

Traditional file storage systems often store data in plain text or use limited protection mechanisms. In such systems, if an attacker gains access to the storage server, the data can easily be viewed, copied, or modified. This creates serious risks for both individuals and organizations. For example, unauthorized access to personal documents may lead to identity theft, while unauthorized access to business data may result in financial losses or damage to an organization's reputation. Therefore, there is a strong need for secure storage systems that protect data from unauthorized access.

One of the most effective solutions to this problem is the use of encryption techniques. Encryption is a process of converting readable data, known as plaintext, into an unreadable format called ciphertext. Only authorized users who possess the correct decryption key can convert the ciphertext back into the original readable data. By using encryption, even if an attacker obtains the stored files, they cannot understand the content without the proper key. Encryption therefore plays a critical role in maintaining data confidentiality and security.

Modern security systems widely use encryption algorithms such as the Advanced Encryption Standard (AES). AES is a symmetric encryption algorithm that uses the same key for both encryption and decryption. It is widely used because of its high level of security and efficiency. AES supports different key sizes such as 128-bit, 192-bit, and 256-bit keys, providing strong protection against brute-force attacks. Many secure applications, including banking

systems, government communication networks, and cloud storage services, rely on AES encryption to protect sensitive data.

Another important aspect of secure data storage is authentication and access control. Authentication ensures that only legitimate users can access the system by verifying their identity through credentials such as usernames and passwords. Access control mechanisms determine what actions each authenticated user is allowed to perform within the system. For example, a user should only be able to access and manage their own files, not the files of other users. By combining authentication and access control with encryption, a system can provide multiple layers of protection against unauthorized access.

Several existing platforms provide online file storage services. Examples include cloud storage services and document management systems that allow users to upload, store, and retrieve files over the internet. While these platforms provide convenience and accessibility, their internal security mechanisms are usually hidden from users. As a result, users may not fully understand how their data is protected. Additionally, such systems are designed mainly for large-scale commercial use rather than educational or experimental purposes.

In the field of computer science education, it is important for students to gain practical knowledge of cybersecurity concepts such as encryption, authentication, and secure data storage. Developing a secure file storage system provides students with an opportunity to understand how these concepts are implemented in real-world applications. By building such a system, students can learn how to integrate cryptographic algorithms, user authentication, and database management into a single secure platform.

The proposed project, titled "**Secure File Storage System Using Encryption and Access Control**," aims to design and develop a system that allows users to securely upload, store, and retrieve files. In this system, all files uploaded by users are encrypted before being stored on the server. This ensures that even if the storage system is compromised, the files remain protected. Only authenticated users can access the system, and each user can only manage their own files. When a user downloads a file, the system decrypts the file and provides the original content to the authorized user.

The system will be implemented using the Python programming language and the Django web framework. Django provides built-in features for user authentication, database management, and secure web development. The system will also use encryption libraries that implement secure cryptographic algorithms such as AES. By combining these technologies, the project will create a secure and efficient file storage platform.

The primary objective of this project is to enhance data security by ensuring that files are stored in an encrypted format and accessed only by authorized users. The system also aims to demonstrate the practical implementation of cybersecurity concepts such as encryption, authentication, and access control. Through this project, users will gain confidence that their files are protected against unauthorized access, while students and developers will gain valuable experience in building secure applications.

In conclusion, the increasing number of cyber threats highlights the importance of secure data storage systems. Encryption-based file storage systems provide an effective solution for protecting sensitive data from unauthorized access. The proposed Secure File Storage System aims to address these challenges by combining strong encryption techniques with authentication and access control mechanisms. This project not only provides a secure method for storing digital files but also contributes to the practical understanding of cybersecurity technologies.

II. Literature Survey

Several researchers and organizations have studied different methods to protect digital data using encryption and secure storage systems. Data security has become a major research area due to the increasing number of cyber threats and unauthorized data access incidents.

One of the most widely used encryption standards is the **Advanced Encryption Standard (AES)**, which was established by the National Institute of Standards and Technology (NIST). AES replaced the older Data Encryption Standard (DES) because DES was vulnerable to brute-force attacks due to its shorter key length. AES provides stronger security and faster performance, making it suitable for protecting large volumes of data. Many secure systems today rely on AES encryption to protect confidential information.

Cloud storage security has also been widely studied by researchers. Many cloud service providers use encryption to protect user data stored on remote servers. According to various studies, encryption helps ensure data confidentiality even if the storage server is compromised. However, some cloud systems rely entirely on provider-controlled encryption, which means users may not have full control over the encryption keys.

Another important area of research focuses on authentication and access control mechanisms. Authentication techniques such as username-password systems, multi-factor authentication, and biometric verification help ensure that only legitimate users can access the system. Access control mechanisms further restrict user actions, ensuring that users can only view or modify resources they are authorized to access.

Several academic projects and studies have also explored how encrypted data storage can be combined with authentication. These systems demonstrate that by integrating cryptographic algorithms, secure user authentication, and database management systems, researchers have shown that it is possible to significantly reduce the risk of data breaches.

The literature indicates that combining encryption with authentication and access control provides a strong foundation for protecting sensitive data. The proposed project builds on these concepts by implementing a secure file storage system that uses encryption techniques to protect user files while allowing controlled access through authenticated user accounts.

TABLE I — Summary of Related Works on Secure File Storage and Encryption

S.No.	Author(s)	Year	Key Contribution	Limitation
1	NIST / Daemen & Rijmen	2001	Introduced AES as the global standard for symmetric encryption replacing DES, supporting 128/192/256-bit keys.	Key management complexity increases with scale; key theft can compromise the entire system.
2	Amazon Web Services (AWS)	2019	Demonstrated server-side AES-256 encryption for cloud object storage, ensuring data-at-rest protection.	Users do not control their own encryption keys; full trust is placed in the service provider.
3	Google Cloud Security Team	2020	Published research on layered encryption and access control for cloud file systems with role-based permissions.	Complex role configurations can introduce misconfigurations leading to unintended access.
4	Khan & Malluhi	2021	Proposed a client-side encryption model where files are encrypted before upload, keeping keys with the user.	Increased client processing load; lost keys result in permanent data loss.
5	Zhang et al.	2022	Explored multi-factor authentication combined with AES encryption in a Django-based secure storage prototype.	Usability concerns for non-technical users; setup complexity for MFA devices.

6	Sharma & Gupta	2023	Developed a role-based access control (RBAC) system for encrypted document sharing in enterprise environments.	RBAC model requires manual role assignment and does not scale automatically with dynamic teams.
7	Ibrahim & Hassan	2023	Examined hybrid encryption combining AES for file content and RSA for key exchange in web-based storage systems.	RSA key exchange adds computational overhead, slowing performance for large file transfers.
8	Patel et al.	2024	Built a Django + Python secure file portal with per-user AES key derivation from login credentials.	If a user changes their password, previously encrypted files may become inaccessible without migration.

III. Methodology

The methodology describes the steps followed to design and develop the Secure File Storage System. The system follows a structured process to ensure that files are securely stored and accessed only by authorized users.

First, the system provides a user registration and login mechanism. Users must create an account and authenticate themselves before accessing the system. This step ensures that only legitimate users can interact with the application.

Second, once the user logs in, they can upload files through the system interface. Before storing the file on the server, the system encrypts the file using a secure encryption algorithm such as AES. The encryption process converts the file into an unreadable format, ensuring that the stored data remains protected.

Third, the encrypted file is stored in the system database or storage directory along with relevant metadata such as the user ID and upload time. This ensures that the system can track file ownership and maintain proper access control.

Fourth, when the user requests to download a file, the system verifies the user's identity and checks whether they have permission to access the requested file. If the user is authorized, the system decrypts the encrypted file and provides the original file to the user.

Finally, the system ensures that each user can only access their own files, preventing unauthorized users from viewing or modifying other users' data. This combination of authentication, encryption, and access control forms the core methodology of the secure file storage system.

A. System Design

The system is designed using the Python programming language and the Django web framework. Django provides built-in tools for managing user authentication, database operations, and URL routing. The system uses a relational database to store user account information, file metadata, and access logs. Encrypted files are stored in a dedicated server-side directory that is not publicly accessible. Only the application layer can retrieve and decrypt files upon authenticated request.

B. Encryption Module

The encryption module uses the Advanced Encryption Standard (AES) algorithm in CBC (Cipher Block Chaining) mode. When a user uploads a file, the system generates a unique encryption key derived from the user's credentials and a server-side secret. The file content is padded to align with AES block size requirements, then encrypted using

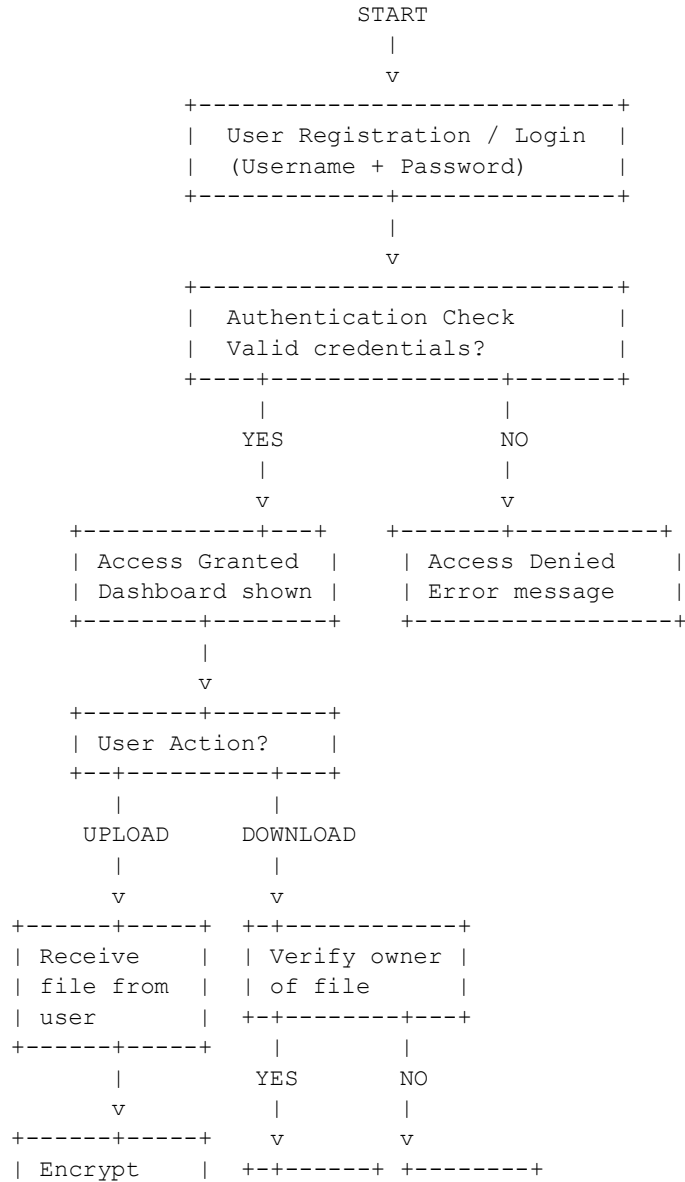
the generated key. The resulting ciphertext is written to the storage directory. The original plaintext file is never written to disk, ensuring that at-rest data is always protected.

C. Authentication and Access Control

User authentication is handled by Django's built-in authentication framework, which securely hashes and stores passwords using the PBKDF2 algorithm with a SHA-256 hash. When a user submits login credentials, the system verifies the provided password against the stored hash without ever storing the plaintext password. Access control is enforced at the view layer: every file retrieval and deletion request is validated to confirm that the requesting user is the file's owner. Requests from users attempting to access files belonging to other accounts are rejected with an appropriate error response.

IV. Methodology Flow Diagrams

A. System Workflow Diagram



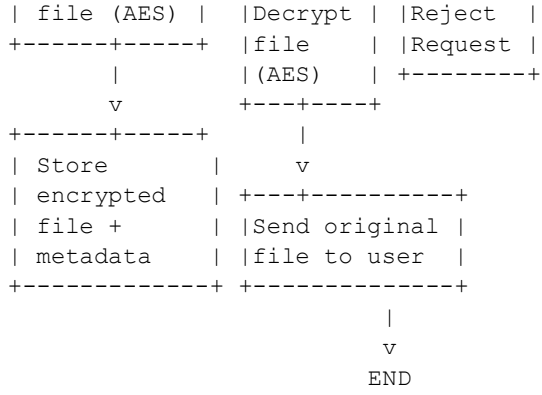


Fig. 1. System workflow diagram — from user login to file upload/download with encryption.

B. Data Flow Diagram (DFD) — Level 0

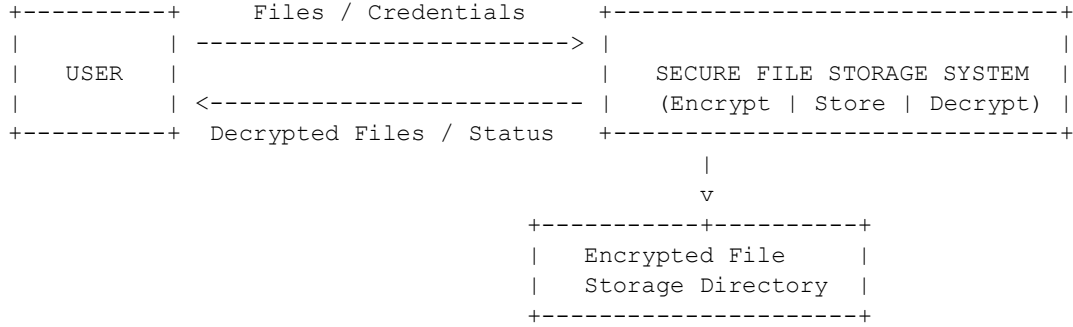
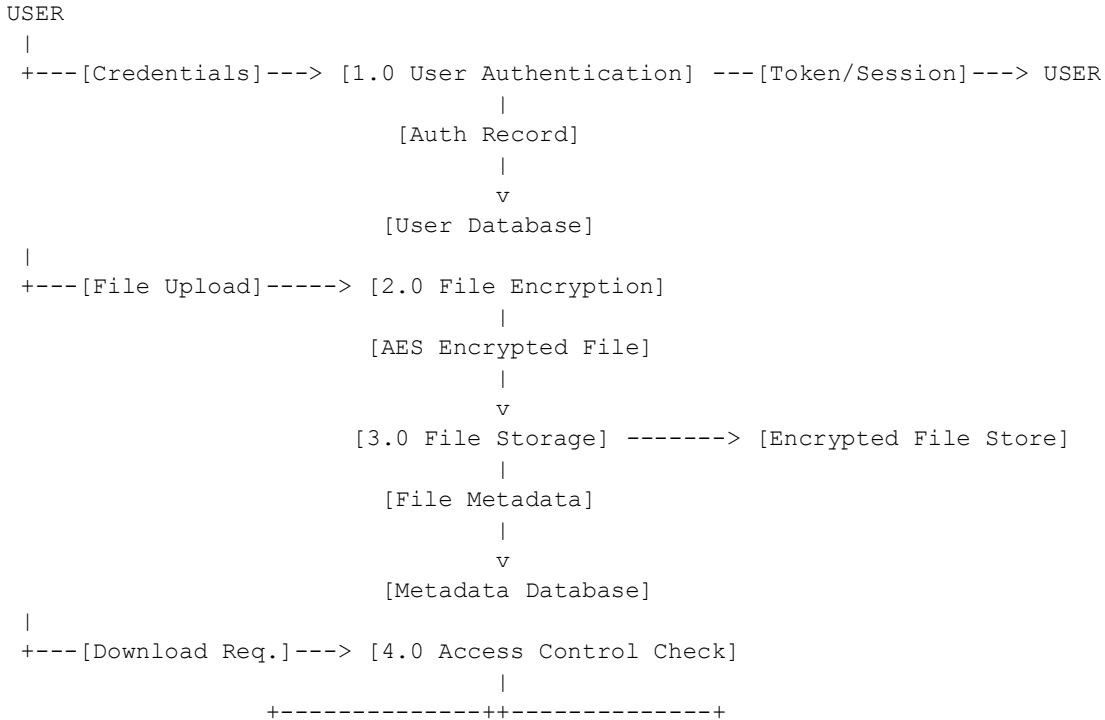


Fig. 2. Level-0 Data Flow Diagram — user interacts with the system; files flow through encryption before storage.

C. Data Flow Diagram (DFD) — Level 1



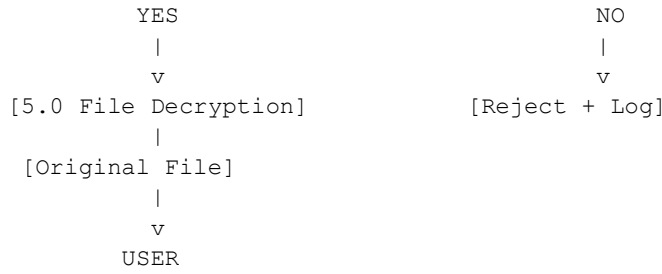


Fig. 3. Level-1 Data Flow Diagram — detailed process flow across authentication, encryption, storage, and decryption.

D. System Architecture Diagram

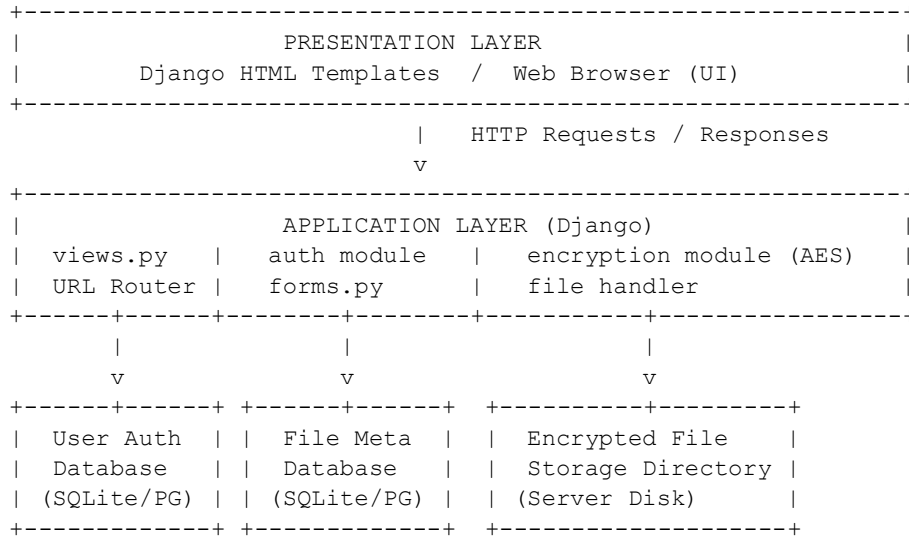


Fig. 4. System architecture — three-layer design separating presentation, application logic, and data storage.

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